Fundamentals of
Bioinformatics-I

VeriCure
BIOLABS PVT. LTD.
Advancing Science, Improving lives.

Dr. Aditya Prasad Panda



The Mission of
VeriCure Biolabs

VeriCure
BIOLABS PVT. LTD.
Advancing Science, Improving lives.

"To empower healthcare and life-science research through innovative, affordable, and accessible molecular diagnostics, bioinformatics, and biotechnology solutions, transforming scientific knowledge into reliable solutions for better health and a healthier future."



VeriCure BioLab

About the Instructor



Dr. Panda is a molecular microbiologist and computational structural biologist with 7+ years of experience in teaching and research. He completed his Ph.D. at IIT Kharagpur, where his research focused on using bioinformatics and computational structural biology to elucidate the molecular mechanisms of β-lactamases and bacterial efflux pump systems involved in antibiotic resistance.

His research combines molecular biology, protein biochemistry, and computational structural biology.

Let’s begin with a

Story!

From Peas to Petabytes…

The Monk and the Peas



Gregor Mendel

Discrete Inheritance Units
Traits pass intact via distinct factors, rejecting parental blending theories.

Quantitative Biology
Counted thousands of pea traits to mathematically infer genetic rules.

34 Years Ignored: Published in 1866, Mendel's work remained unrecognized until its rediscovery in 1900.

Unmasking the Hereditary Molecule

1869 | **Miescher Isolates Nuclein**
Phosphorus-rich DNA extracted from white blood cell nuclei.

1944 | **Avery-MacLeod-McCarty**
Proves DNA transforms bacterial traits, overturning protein bias.

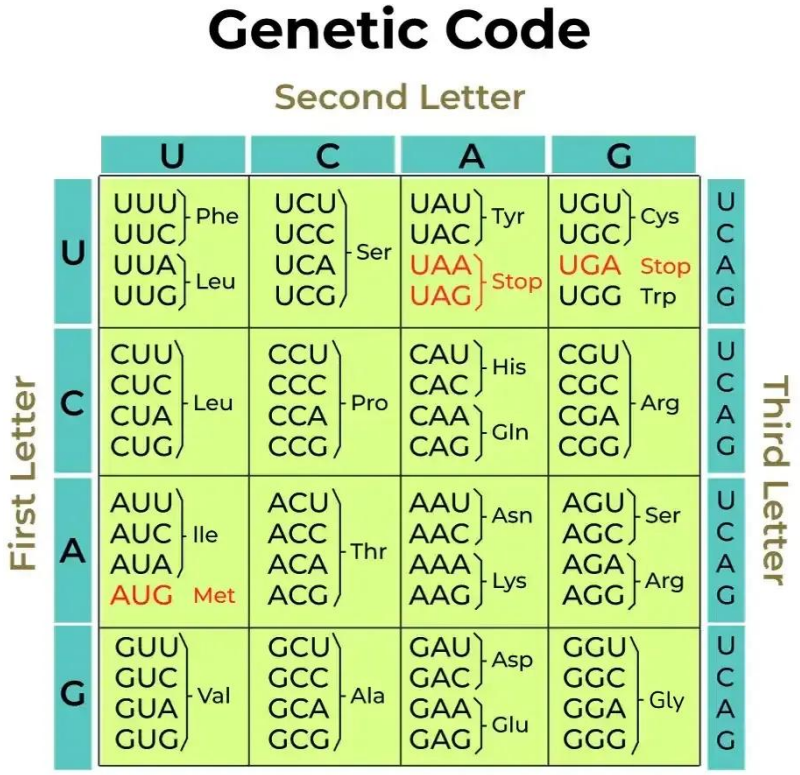
1952 | **Rosalind Franklin's Photo 51**
X-ray diffraction captures the definitive helical signature.

1953 | **Watson & Crick Double Helix**
3D model directly reveals the biological copying mechanism.



Rosalind Franklin

The Genetic Code and the Birth of rDNA Technology



1958 · Central Dogma

Crick dictates the directional flow of information: **DNA → RNA → Protein**.

1961 · First Codon Deciphered

Nirenberg & Matthaei match **UUU to Phenylalanine**, unlocking 64 codons for 20 amino acids.

1970s · Recombinant DNA

Bacterial restriction enzymes allow scientists to cut, paste, and edit genes across species.

Reading the Code &

The Computer Shift!

5,386

Base Pairs Sequenced in Phage ΦX174

Frederick Sanger's 1977 landmark method achieved the first complete genome.

The Computational Turning Point

Fragment assembly by hand proved impossible. Computers entered biology out of sheer necessity to align overlapping reads.



Frederick Sanger

The Birth of

Bioinformatics Tools



1965 PIONEER

Margaret Dayhoff

Mother of Bioinformatics. Published the 1st Protein Atlas & PAM substitution matrices.

1971 | **Protein Data Bank (PDB)**
3D structural archive launched with 7 initial atomic structures.

1982 | **GenBank Sequence Archive**
A universal open repository has been established for DNA sequence data.

1990 | **BLAST Alignment Tool**
An algorithm enabling fast similarity searching across millions of sequences.

Global Megaproject (1990–2003)

Human Genome Project

3 Billion
Chemical Base Pairs Decoded

$2.7B
13-Year Project Cost

< $500
Genome Cost Today (<1 Day)

Public Consortium vs. Celera Race

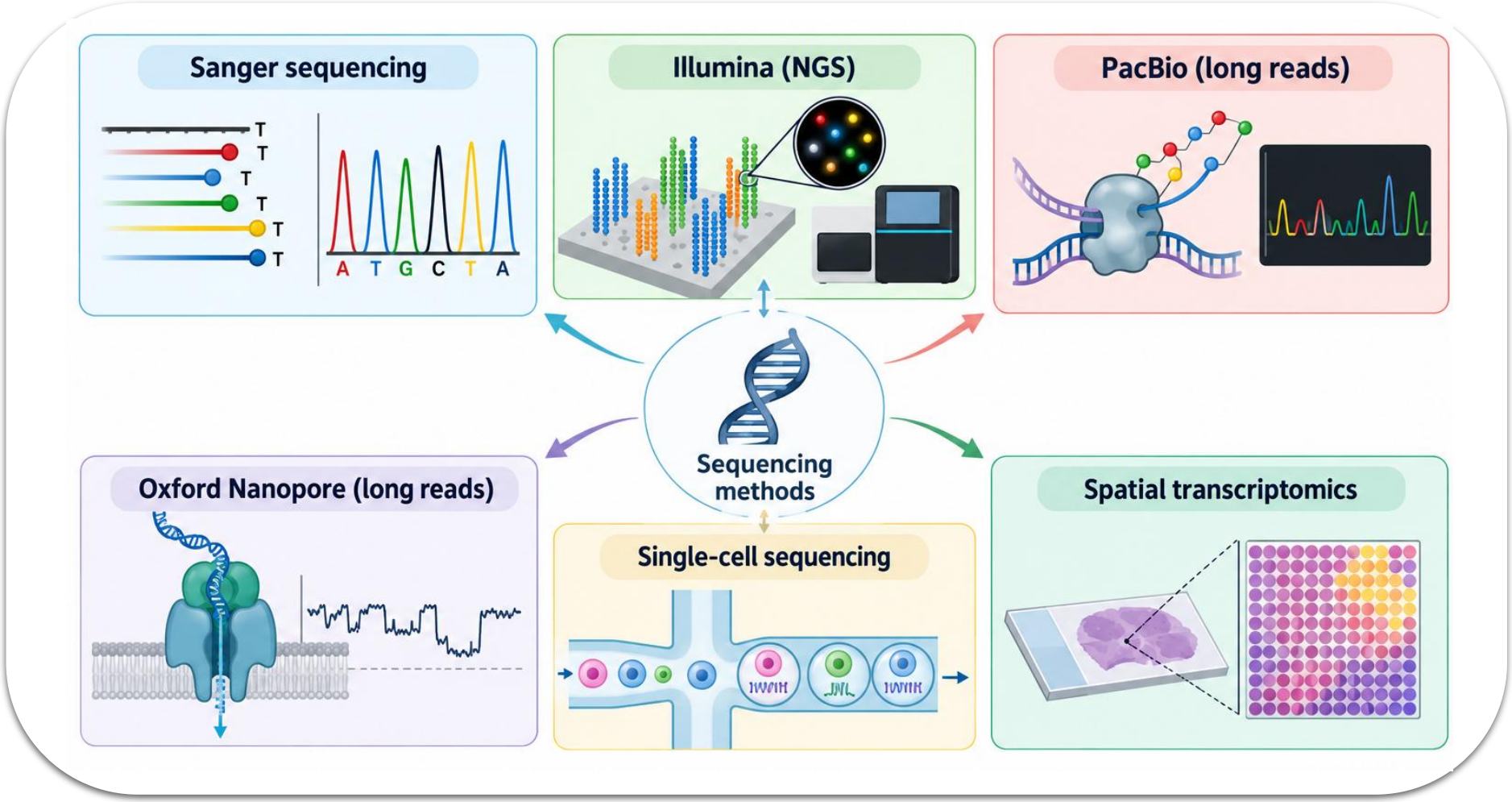
Shotgun sequencing algorithms accelerated computational assembly.

2022 Telomere-to-Telomere (T2T)

Resolved the final 8% repetitive gaps for the first truly gapless sequence.

The growth in

Sequencing technology



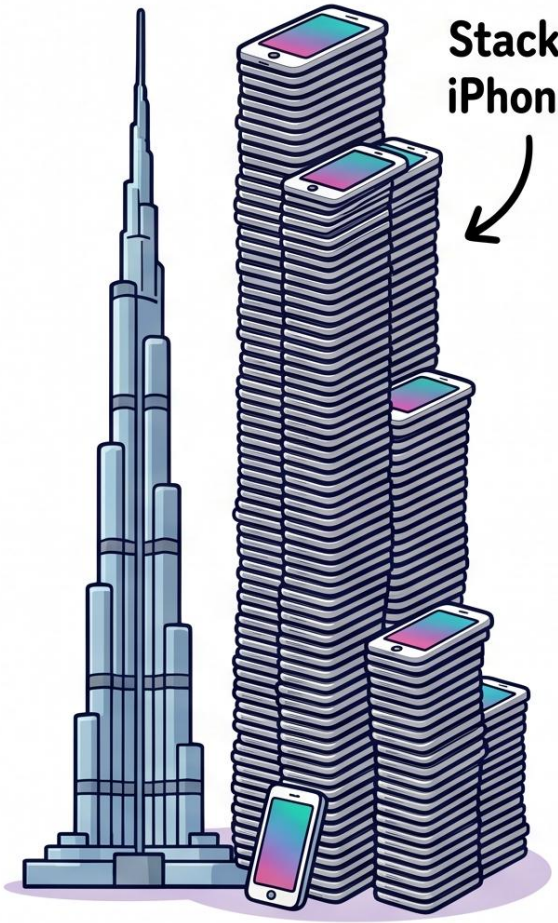
The growth in

Sequencing technology…

The BIG-DATA Problem!!

100 PETABYTES of DNA Sequence Data

How big is 100 Petabytes (PB)?



Stack of 100,000 iPhones!
(Taller than the Burj Khalifa)



3,800 YEARS of
continuous 4K video!



A paper highway of books
reaching to the Moon & back!

How much DNA data is generated daily?

Average Person:
~2 GB/day
(400 photos or 1hr video)



VS

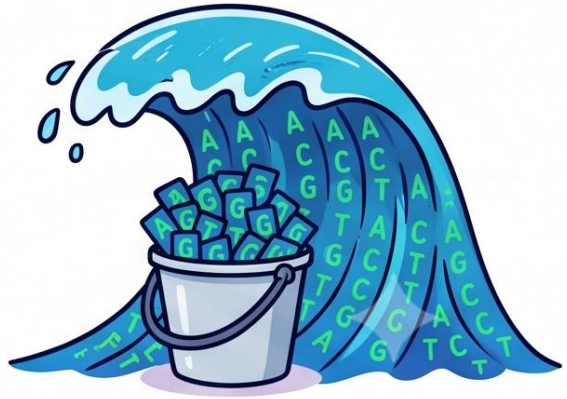
1 DNA Sequencer:
Up to 16,000 GB/day
(up to 300 human genomes!)





Every video on
YouTube in a year

VS



Global DNA Sequence Data
generated in just a few weeks

How on Earth do we store and read

100 Petabytes of DNA?

Store in a **DATABASE SERVER!**

What is a

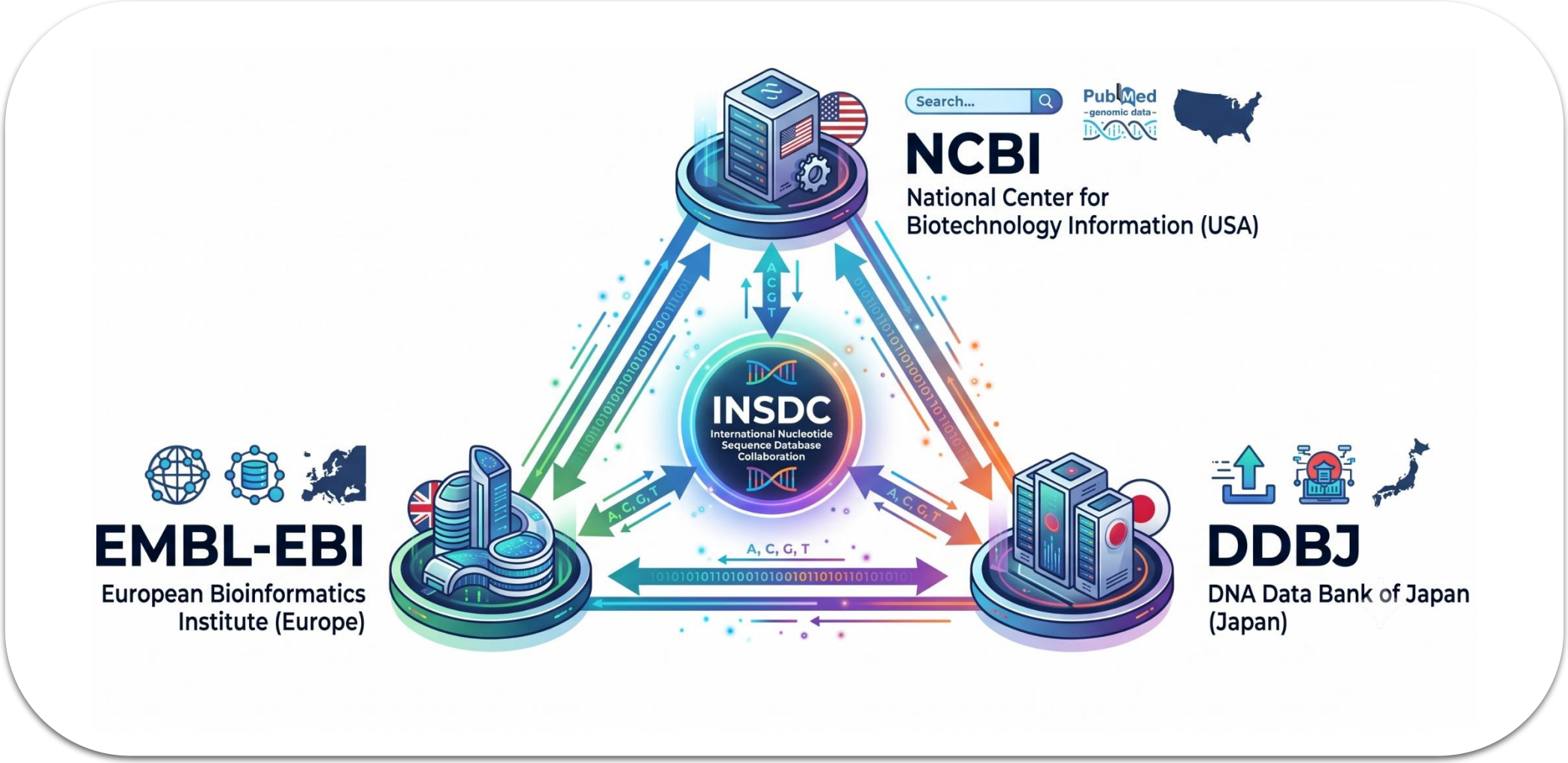
Database server?

“A database server is a dedicated computer system or software application that stores, manages, and provides database services (such as data retrieval, security, and updates) to other programs or computers via a client-server architecture



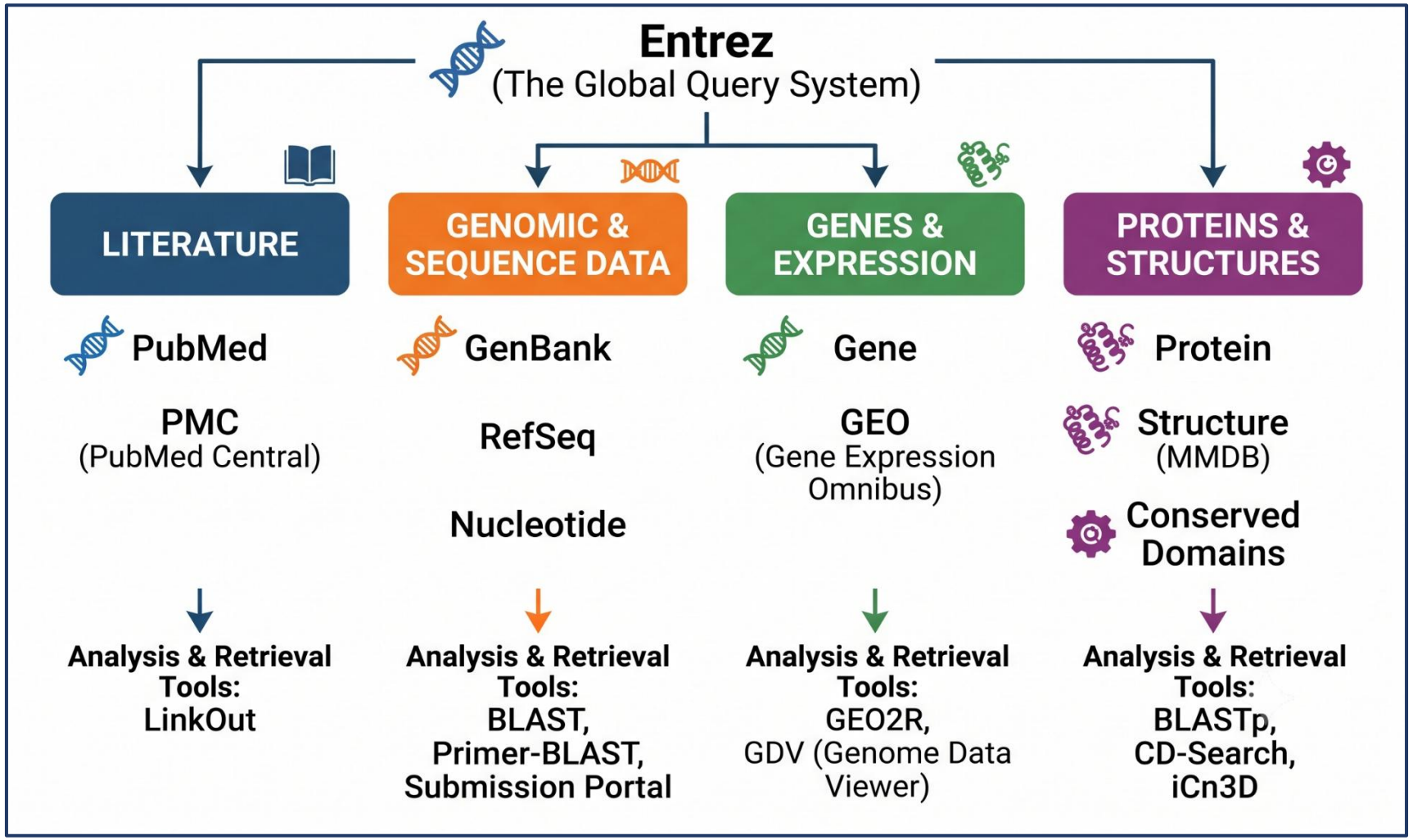
Global co-ordination

The INSDC Alliance!



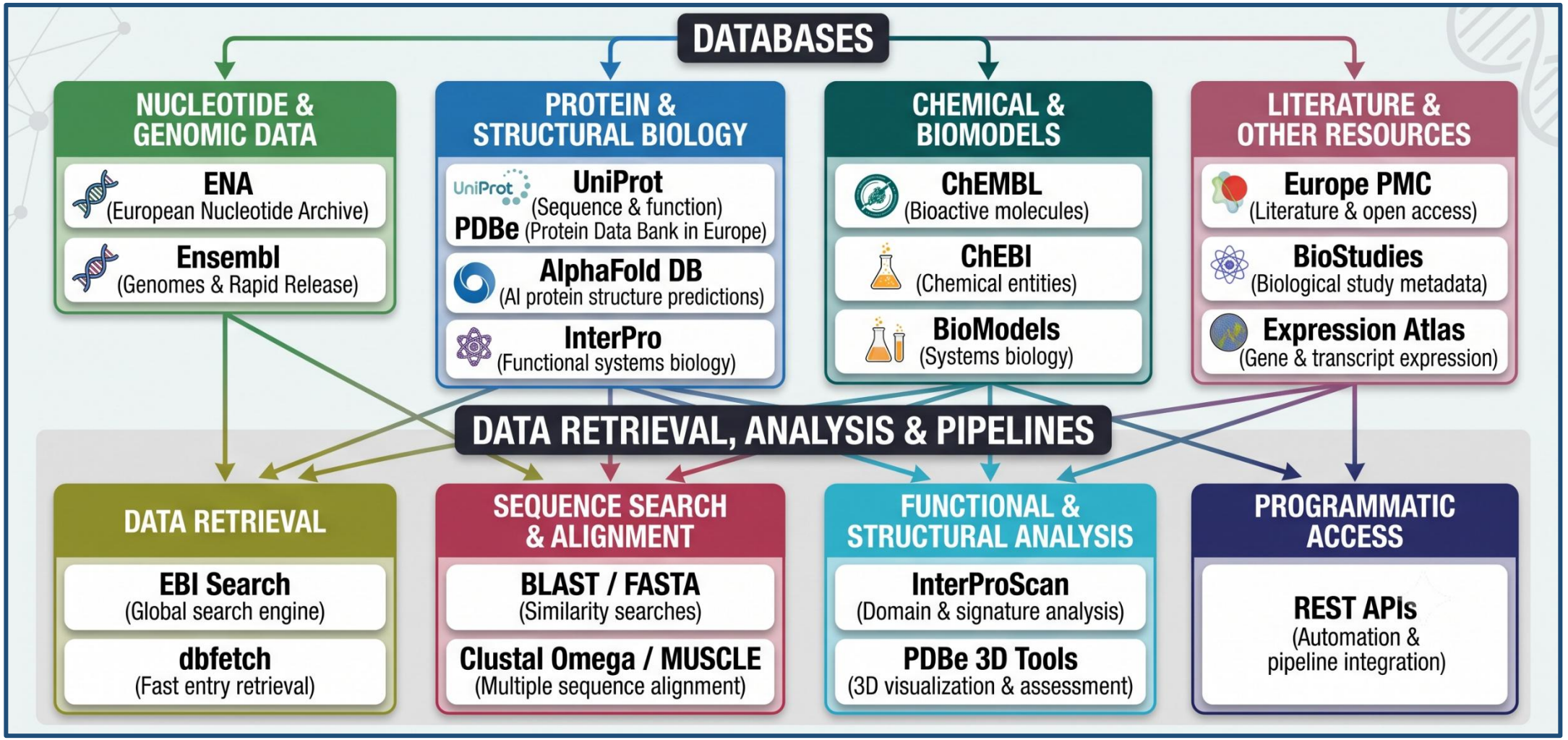
Major Databases & Tools

NCBI Ecosystem



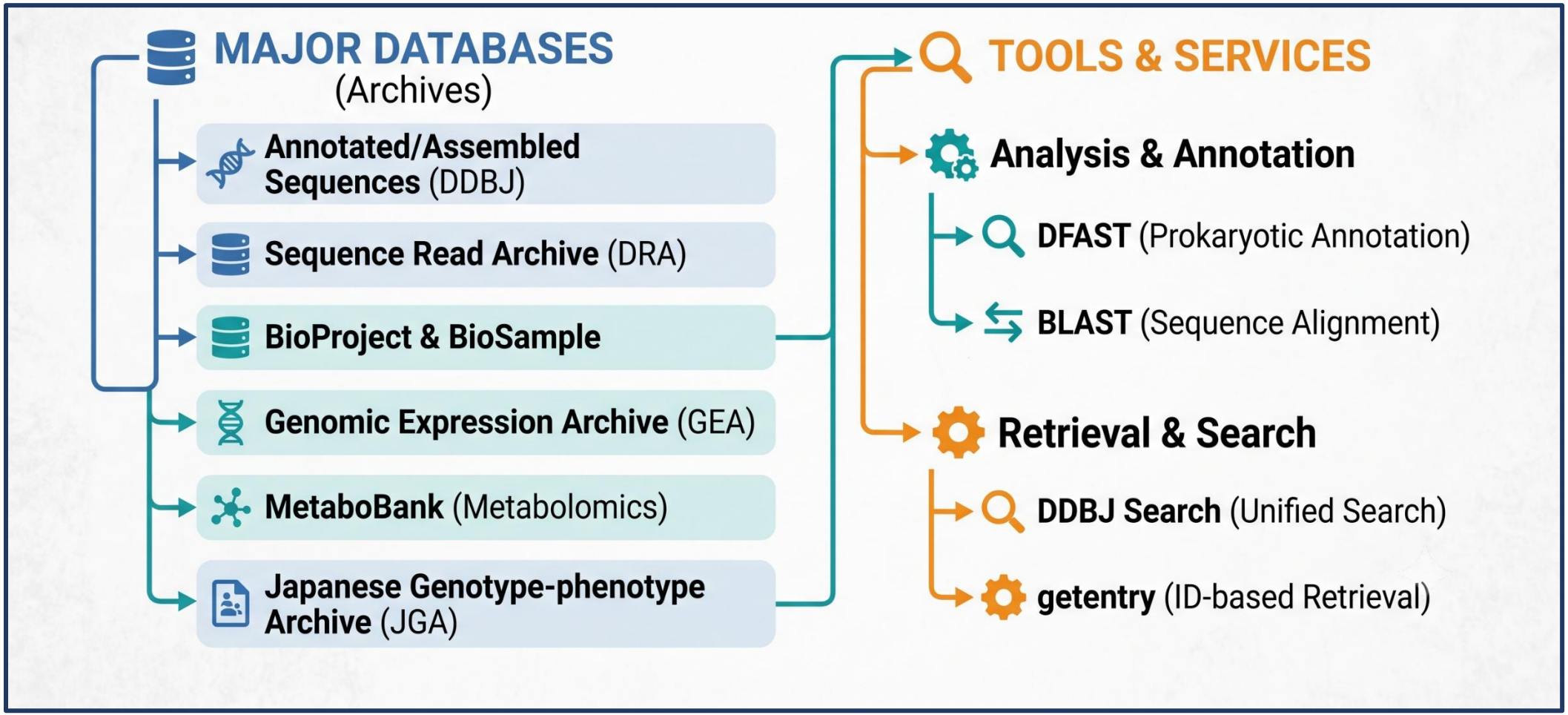
Major Databases & Tools

EMBL-EBI



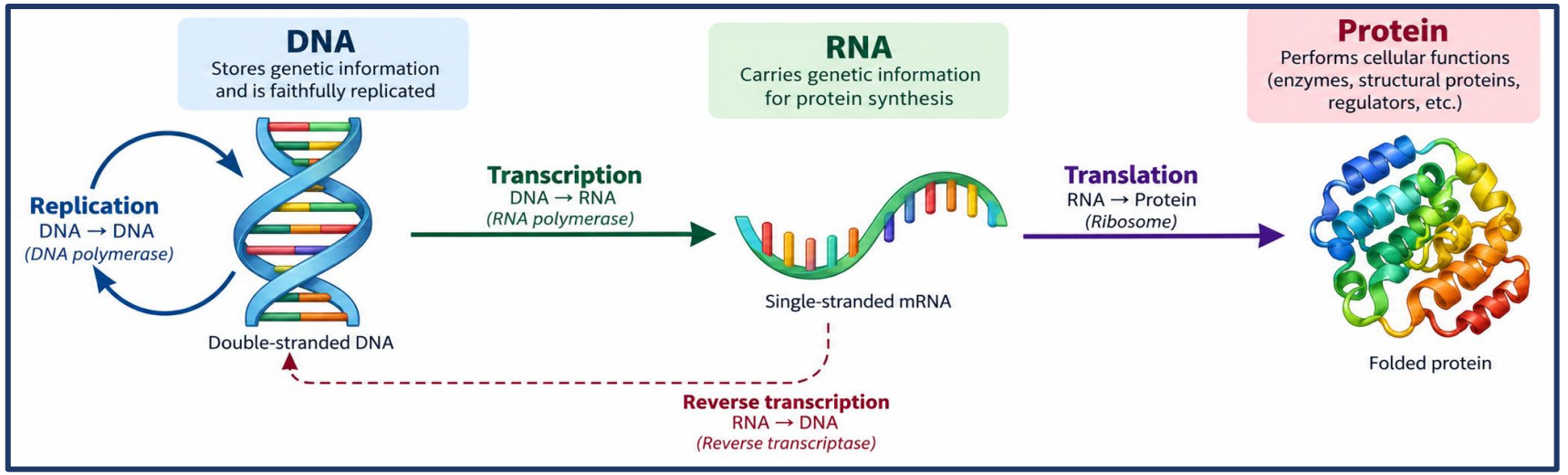
Major Databases & Tools

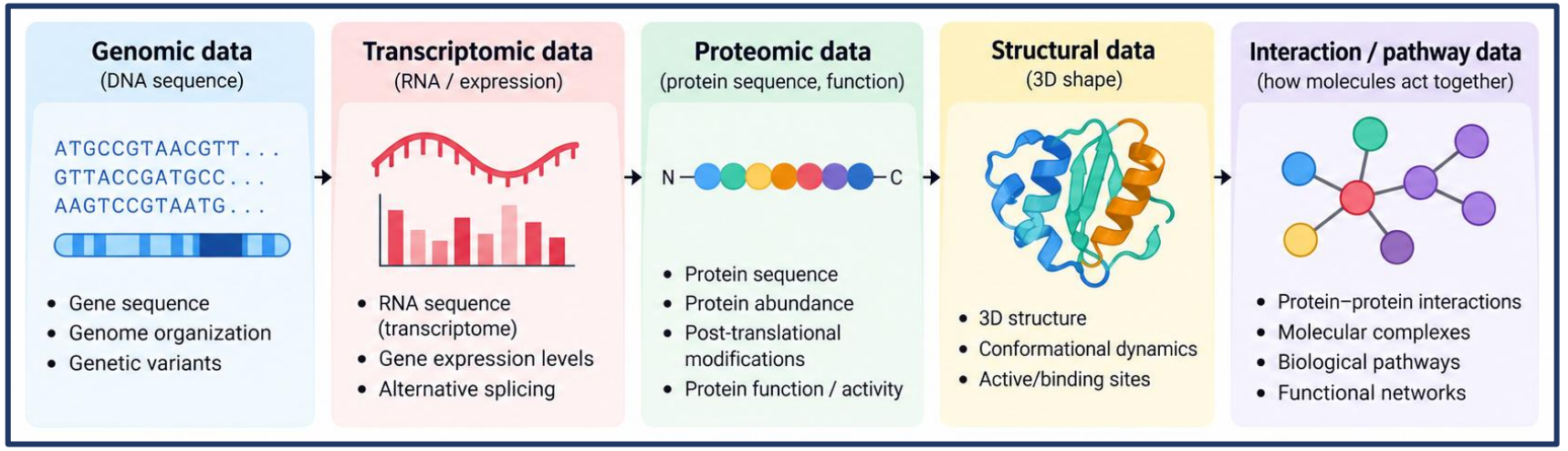
DDBJ



Biology Primer

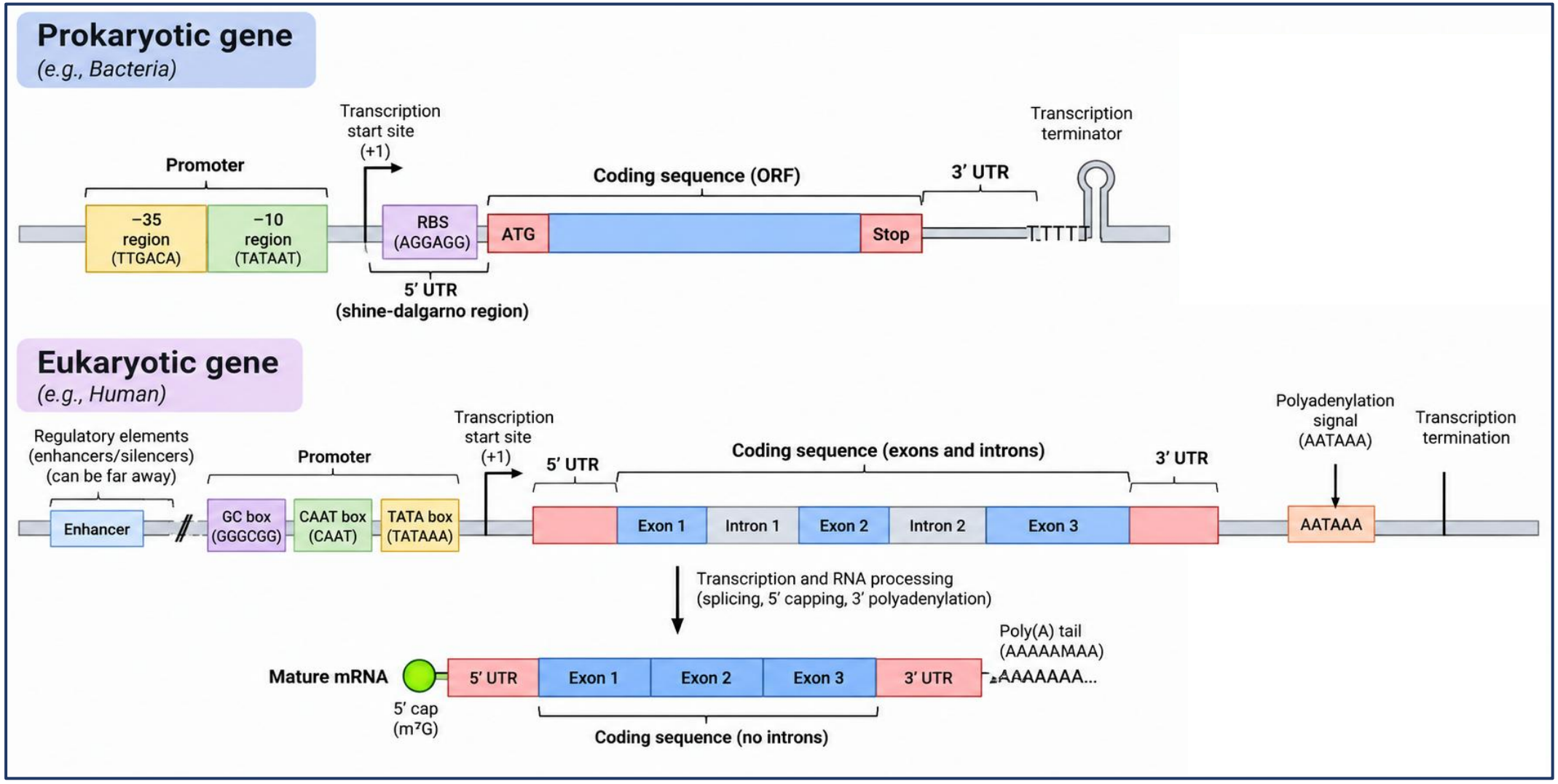
Central Dogma of Molecular Biology





Biology Primer

Gene Structure



Biology Primer

Alternative splicing

